REVISION 2026 | Semester I | Practicum Course | Complete Handbook

Introduction to Printing Technology

Prepress • Press • Postpress • Screen Printing Operations • Laboratory Safety & Discipline

POLY PMNA

Course Code	**1101**
Programme	**Printing Technology**
Contact Hours	**90**
L:T:P:S	**3:0:3:0**
Self-Learning (SS)	**6 hrs/week**
Credits	**4.5**
Prerequisites	**NIL**
Bloom Level	**Apply**

Practicum Focus: Prepress design/typesetting, press ink-transfer, postpress finishing, and single/multi-colour screen printing with registration safety.

OFFICIAL SOURCE DECLARATION & TRANSPARENCY

Verbatim SITTTR Syllabus Data & Disclosed Discrepancies

This handbook is compiled strictly from the official SITTTR Kerala Diploma Revision 2026 Course Description page for Subject Code 1101. To ensure 100% academic honesty, all verified inconsistencies in the official source are explicitly declared below.

1. Conflicting Assessment Scheme Values in Source

The official syllabus document contains two conflicting evaluation tables:

- **Top "COURSE DESCRIPTION" Table:** States `CIE Marks = 40`, `ESE Marks = 100` (Total 140).
- **Bottom "Teaching and Assessment Scheme" Table:** Itemises `Theory: CIE 20 + ESE 60 = 80`, `Practical: CIE 20 + ESE 40 = 60` (Grand Total = 140).

Disclosed Status: Both tables have been reproduced verbatim in this handbook. Students and faculty should consult SITTTR board circulars regarding the internal/external mark distribution for evaluation.

2. Verified Typographical Value in CO–PO Matrix

In the official CO–PO Articulation Matrix, the **Correlation Level** row for column P03 is printed exactly as `1,5` (using a comma).

Disclosed Status: This value has been preserved as `1,5` exactly as printed in the official source data, flagged as a probable typographical error for `1.5`.

⚠ **Missing Source Data Disclosure:** The supplied official syllabus PDF for 1101 contains only the front course-description page. It does *NOT* contain module titles, module-wise topic lists, module hour allocations, official practical job lists, prescribed textbooks, or references. As required by quality rules, these sections are explicitly marked **"Not specified in the supplied official syllabus PDF — pending module/practical/reference pages"** and have not been fabricated.

COURSE DASHBOARD

Course Structure and Assessment Scheme

Subject 1101 introduces printing technology, equipment, workflow stages, and screen printing practical operations in Semester I of the Diploma in Printing Technology programme.

90
Contact hours

4.5
Credits

3:0:3:0
L : T : P : S scheme

Practicum
Course type

Teaching and Assessment Scheme (Verbatim SITTTR Table)						
Teaching Scheme / Week	**Self-Learning (SS) / Week**	**Total Hours / Semester**	**Credits (C)**	**Examination Scheme (Theory)**	**Examination Scheme (Practical)**	**Grand Total**
L = 3, T = 0, P = 3, S = 0	6 Hours	90 Hours	4.5	CIE: 20 | ESE: 60 (Total: 80)	CIE: 20 | ESE: 40 (Total: 60)	140 Marks

Footnote (verbatim): L: Lecture (1 hr = 1 credit), T: Tutorial (1 hr = 1 credit), P: Practical (1 hr = 0.5 credit), S: Project (1 hr = 0.5 credit), SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination.

Course Prerequisites			
Topic	**Course Code**	**Course Title**	**Semester**
NIL	(blank)	(blank)	(blank)

OBJECTIVES & COURSE OUTCOMES

Curriculum Goals and Competency Mapping

Course Objectives (Verbatim)

1. Understand the role and importance of printing in daily life
2. Demonstrate basic printing workflow activities (prepress, press, postpress)
3. Identify basic printing processes, machines, and materials
4. Identify fundamental printing and finishing equipment and its components
5. Follow safe and disciplined laboratory practices
6. Perform basic screen printing operations

Course Outcomes (Verbatim)

- **CO1:** Explain the role and evolution of printing in society and industry — *Bloom's Level: Understand*
- **CO2:** Demonstrate basic prepress, press and postpress workflow activities — *Bloom's Level: Apply*
- **CO3:** Identify printing processes and Practice laboratory safety, discipline, and teamwork — *Bloom's Level: Apply*
- **CO4:** Perform single and multi-colour screen printing with proper registration and safety practices — *Bloom's Level: Apply*

CO–PO Mapping: Course Articulation Matrix (Verbatim)

CO–PO Articulation Matrix											
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1	2	1	-	-	-	1	-	-	-	1	1
CO2	2	2	1	1	1	-	-	-	1	1	-
CO3	2	1	-	-	1	1	1	2	2	-	-
CO4	2	2	2	1	1	-	-	1	1	-	-
Total	**8**	**6**	**3**	**2**	**3**	**2**	**1**	**3**	**4**	**2**	**1**
Correlation Level	**2**	**1.5**	**1,5***	**1**	**1**	**1**	**1**	**1.5**	**1.33**	**1**	**1**

Legend: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-".
Note: Value 1,5 for PO3 is reproduced exactly as printed in the official SITTTR syllabus document (probable typo for 1.5).

COVERAGE MATRIX

Internal Syllabus Coverage Status

Confirmed Syllabus Components vs Pending Content

PDF Source Section	Component	Official Topic / Data	Hours	Target CO	Bloom Level	Handbook Section Status
Course Info	Subject Identity	1101 — Introduction to Printing Technology	90	CO1–CO4	Understand / Apply	Fully Populated (Cover & Dashboard)
Introduction	Course Scope	Role of printing, prepress/press/ postpress, lab safety	90	CO1–CO4	Understand / Apply	Fully Populated (Section 1–4)
Objectives	Objectives 1–6	Role in daily life, workflow, processes, safety, screen printing	90	CO1–CO4	Understand / Apply	Fully Populated (Section 4)
Outcomes	CO1–CO4	Role/evolution, workflow, processes/safety, screen printing	90	CO1–CO4	Understand / Apply	Fully Populated (Section 4)
Assessment	Scheme Tables	CIE 40 / ESE 100 vs Theory 80 / Practical 60	90	CO1–CO4	Apply	Fully Populated & Disclosed
Module Pages	Module 1 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO1	Understand	Pending Notice Inserted (Module 1)
Module Pages	Module 2 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO2	Apply	Pending Notice Inserted (Module 2)
Module Pages	Module 3 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO3	Apply	Pending Notice Inserted (Module 3)
Module Pages	Module 4 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO4	Apply	Pending Notice Inserted (Module 4)
Practical Pages	Experiment Job List	Pending — awaiting official screen-printing practical job list	45 (P)	CO4	Apply	Pending Notice Inserted (Section 8 & 11)

LEARNING ROADMAP

Provisional Workflow Progression Overview

MODULE CHAPTERS

Syllabus Content & Fundamental Orientation

MODULE I (THEMED ORIENTATION)

Role and Evolution of Printing in Society and Industry

Understanding how printing impacts daily life, education, packaging, advertising, logistics, and mass media.

CO1 Target | Bloom: Understand | Theory / Orientation

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: Exact module unit numbers, topic sub-headings, and hour allocations for Module I will be updated when the remaining syllabus pages are provided. The orientation below is built upon confirmed Course Objective 1 and Outcome CO1.

1.1 Importance of Printing in Daily Life and Communication–

Printing is a fundamental communication medium that enables the mass reproduction of text, graphics, and visual images. In daily life, printed products appear in education (textbooks, notebooks), packaging (box cartons, food wraps, labels), advertising (posters, flyers, banners), logistics (barcodes, shipping labels, waybills), and media (newspapers, magazines).

Malayalam explanation: അച്ചടി (Printing) മനുഷ്യജീവിതത്തിലും ആശയവിനിമയത്തിലും പ്രധാന പങ്കുവഹിക്കുന്നു. വിദ്യാഭ്യാസം (പുസ്തകങ്ങൾ), പാക്കേജിംഗ് (ലേബലുകൾ, ബോക്സുകൾ), പരസ്യം (പോസ്റ്ററുകൾ, ഫ്ലയറുകൾ), ലോജിസ്റ്റിക്സ് (ബാറുകൾ, ലേബലുകൾ) എന്നിവയിലെല്ലാം പ്രിന്റിംഗ് അത്യന്താപേക്ഷിതമാണ്. Technical terms: *Packaging, Advertising, Logistics, Media.*

Industry Relevance: Modern printing integrates digital prepress with high-speed production presses to supply essential visual communication tools worldwide.

1.2 Evolution of Printing Technologies–

Printing evolved from manual woodblock carving and Gutenberg's movable metal type (1440s) to industrial letterpress, planographic offset printing, gravure, screen printing, and modern digital/variable-data inkjet printing systems.

Malayalam explanation: മാനുവൽ വുഡ് ബ്ലോക്ക് പ്രിന്റിംഗിൽ നിന്ന് ആരംഭിച്ച് ഗുട്ടൻബെർഗിന്റെ മൂവബിൾ ടൈപ്പ്, ലെറ്റർപ്രസ്സ്, ഓഫ്സെറ്റ് പ്രിന്റിംഗ്, ഗ്രാവിവർ, സ്ക്രീൻ പ്രിന്റിംഗ്, ഡിജിറ്റൽ ഇങ്ക്ജെറ്റ് ടെക്നോളജി എന്നിവയിലേക്ക് അച്ചടി വികസിച്ചു.

MODULE II (THEMED ORIENTATION)

Basic Printing Workflow Activities (Prepress, Press & Postpress)

Demonstrating the three fundamental phases of print manufacturing from initial design to finished product.

CO2 Target | Bloom: Apply | Workflow Sequence

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: Detailed topic list and hour allocations for Module II are pending syllabus page release. Content below reflects confirmed CO2 and Objective 2.

2.1 Prepress Workflow (Design, Layout & Plate Making)−

Prepress encompasses all activities prior to actual printing on press. Key steps include graphic design, image capture, typesetting, colour separation (CMYK), page imposition, computer-to-plate (CTP) imaging or stencil preparation for screen printing.

> **Malayalam explanation:** പ്രിന്റിംഗിന് മുൻപുള്ള പ്രവർത്തനങ്ങളാണ് Prepress. ഡിസൈനിംഗ്, ഇമേജ് അസംബ്ലി, കളർ സെപ്പറേഷൻ (CMYK), ഇമ്പോസിഷൻ, പ്ലേറ്റ് മേക്കിംഗ് / സ്റ്റെൻസിൽ തയ്യാറാക്കൽ എന്നിവ ഇതിൽ ഉൾപ്പെടുന്നു.

2.2 Press Operations (Ink Transfer & Printing Systems)−

The Press stage involves transferring ink from an image carrier (plate, cylinder, screen) onto a substrate (paper, plastic, fabric, metal). It includes sheet feeding, ink metering, damping (in offset), cylinder impression, and register control.

> **Malayalam explanation:** പ്ലേറ്റിൽ നിന്നോ സ്ക്രീനിൽ നിന്നോ മഷി കാർഡിലേക്കോ പേപ്പറിലേക്കോ പകർത്തുന്ന ഘട്ടമാണ് Press stage. ഇങ്ക് ഡോസിംഗ്, ഇംപ്രഷൻ മർദ്ദം, രജിസ്ട്രേഷൻ കൺട്രോൾ എന്നിവ പ്രധാനമാണ്.

2.3 Postpress Operations (Finishing & Binding)−

Postpress includes all finishing operations performed after printing: guillotine cutting, folding, collating, stitching/binding, trimming, lamination, die-cutting, and foil stamping to produce final books, boxes, or cards.

> **Malayalam explanation:** പ്രിന്റിംഗിന് ശേഷമുള്ള പ്രക്രിയകളാണ് Postpress (Finishing). കട്ടിംഗ്, ഫോൾഡിംഗ്, സ്റ്റിച്ചിംഗ്/ബൈൻഡിംഗ്, ട്രിമ്മിംഗ്, ലാമിനേഷൻ എന്നിവ ഇതിൽപ്പെടുന്നു.

MODULE III (THEMED ORIENTATION)

Printing Processes, Machines, Materials & Laboratory Safety

Identification of major printing processes and strict enforcement of laboratory safety, discipline, and teamwork.

CO3 Target | Bloom: Apply | Processes & Safety

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: Topic details for Module III are pending syllabus expansion. Content below reflects confirmed CO3, Objectives 3, 4 & 5.

3.1 Major Printing Processes & Image Carriers–

Major Printing Processes Comparison

Process	Image Carrier Surface	Ink Transfer Principle	Typical Products
Relief (Flexo / Letterpress)	Raised image areas	Ink applied to raised surface and pressed onto substrate	Labels, corrugated boxes, packaging
Planographic (Offset)	Flat surface (Chemical balance)	Ink accepts oil-receptive image; water repels non-image	Textbooks, newspapers, magazines
Intaglio (Gravure)	Sunken / Recessed cells	Ink filled in cells; excess wiped by doctor blade	High-volume packaging, currency
Stencil (Screen Printing)	Porouse mesh screen	Ink squeezed through open mesh areas with a squeegee	Textiles, T-shirts, signboards, circuit boards

Malayalam explanation: നാല് പ്രധാന അച്ചടി രീതികൾ: Relief (ഉയർന്ന ഉപരിതലം), Planographic/Offset (സമനിരപ്പായ ഉപരിതലം), Intaglio/Gravure (ആഴത്തിലുള്ള സെല്ലുകൾ), Stencil/Screen Printing (തുളയുള്ള മെഷ് സ്ക്രീൻ).

3.2 Laboratory Safety Rules, Discipline & Teamwork–

Mandatory Laboratory Safety Rules:

- Wear protective lab aprons and closed leather footwear at all times.
- Keep long hair tied back; remove loose jewellery, ties, and hanging clothing near rotating press rollers.
- Ensure proper room ventilation when handling solvents, wash-up chemicals, and photo emulsions.
- Know the exact location of Emergency Stop buttons, eyewash stations, and fire extinguishers.
- Maintain discipline, cleanliness, and active teamwork during practical lab exercises.

Malayalam explanation: ലബോറട്ടറി സുരക്ഷാ നിയമങ്ങൾ: ലാബ് കോട്ട്, സേഫ്റ്റി ഷൂസ് ധരിക്കുക. കറങ്ങുന്ന മെഷീനുകൾക്കടുത്ത് അയഞ്ഞ വസ്ത്രങ്ങൾ ഒഴിവാക്കുക. കെമിക്കലുകൾ കൈകാര്യം ചെയ്യുമ്പോൾ വായുസഞ്ചാരം ഉറപ്പാക്കുക. അടിയന്തര സ്റ്റോപ്പ് ബട്ടൺ, അഗ്നിശമന ഉപകരണം എന്നിവ മനസ്സിലാക്കുക.

MODULE IV (THEMED ORIENTATION)

Single & Multi-Colour Screen Printing Operations

Practical stencil preparation, squeegee handling, registration techniques, and safety practices.

CO4 Target | Bloom: Apply | Screen Printing Lab

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: Practical experiment breakdown for Module IV is pending official syllabus job list. Content below is based on confirmed CO4 & Objective 6.

4.1 Screen Printing Frame, Mesh, Emulsion & Stencil Preparation−

Screen printing utilizes a wooden or aluminum frame stretched with synthetic polyester mesh. A photo-polymer emulsion is coated on the mesh, dried, exposed to UV light under a film positive, and washed with water to open the image stencil area.

> **Malayalam explanation:** സ്ക്രീൻ പ്രിന്റിംഗ് നിർമ്മാണം: ഫ്രെയിമിൽ പോളിയെസ്റ്റർ മെഷ് വലിച്ചുകെട്ടുന്നു. ഇതിൽ എമൽഷൻ തേച്ചുണക്കി ഫിലിം പോസിറ്റീവ് ഉപയോഗിച്ച് പ്രകാശം അടിപ്പിച്ച് വെള്ളത്തിൽ കഴുകി സ്ക്രീൻ സ്റ്റെൻസിൽ തയ്യാറാക്കുന്നു. Technical terms: *Frame, Mesh, Emulsion, Exposure, Stencil.*

4.2 Registration Controls & Multi-Colour Screen Printing−

Registration ensures exact alignment of images, especially during multi-colour printing. Mechanical stops or registration pins hold the substrate in the exact same position for each successive colour pass.

1. Prepare separate film positives and exposed screens for each colour.
2. Fix registration lay-guides on the printing bed.
3. Print the first colour, allow to dry completely.
4. Align the second screen with registration marks and print the second colour.

> **Malayalam explanation:** മൾട്ടി-കളർ പ്രിന്റിംഗിൽ ഓരോ നിറവും കൃത്യമായ സ്ഥാനത്ത് അച്ചടിക്കാൻ Registration മാർക്കുകളും ഫിക്സഡ് ഗൈഡുകളും ഉപയോഗിക്കുന്നു.

PROCEDURE BANK

Standard Screen Printing Lab Operation Procedure

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: The specific experiment job titles and detailed step lists will be updated upon receipt of the official SITTTR practical manual. The standardized procedure below provides the essential operational foundation for CO4.

Standard Operating Procedure: Single-Colour Screen Printing

Aim: To prepare a screen stencil, set up registration guides, and perform single-colour screen printing on paper/cardboard substrate with proper safety practices.

Apparatus & Materials: Screen frame with polyester mesh, photo emulsion, sensitizer, UV exposure unit, water spray gun, squeegee (70° shore hardness), screen printing ink, substrate, apron, gloves.

1. **Mesh Degreasing & Coating:**

Clean the mesh with degreaser. Apply photo emulsion evenly on both sides using a scoop coater in yellow/safe light.

2. **Drying & Exposure:**

Dry the coated screen horizontally in a dark drying cabinet. Place the film positive on the glass bed and expose to UV light for the calculated duration.

3 **Development & Wash-out:**

Spray cold water gently to wash away unexposed emulsion, revealing the open mesh stencil. Dry the screen completely.

4 **Screen Setup & Registration:**

Clamp the screen in the printing bed. Position lay-guides on the bed so the substrate aligns precisely under the stencil.

5 **Ink Application & Squeegee Pass:**

Apply ink along the top border of the screen. Hold the squeegee at a 45° to 60° angle and draw across with uniform downward pressure.

6 **Drying & Reclaiming:**

Place printed sheets on drying racks. Clean remaining ink from screen using solvent and wash with reclaiming chemical.

Result: Uniform, sharp screen print obtained with crisp edges and proper ink coverage.

INTERACTIVE PRINTING LABORATORY

Diagrams, Flowcharts, Animations and Course Calculators

Prepress → Press → Postpress Workflow

Figure 1: The three major sequential stages of print manufacturing.

Screen Printing Setup & Component Identification

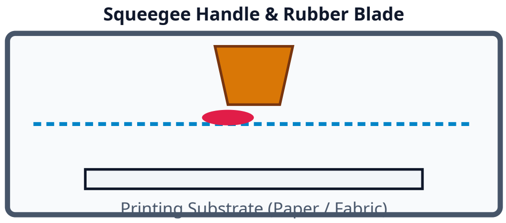
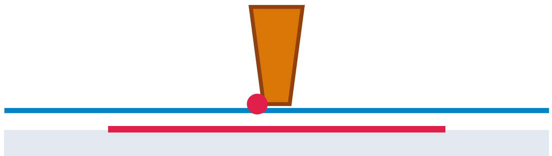


Figure 2: Component identification of screen frame, mesh, ink, squeegee, and substrate.

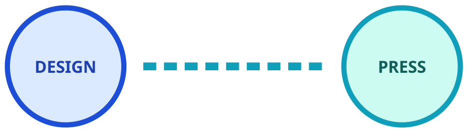
Interactive Animations

Screen Printing Ink Stroke Animation



Demonstrates squeegee action forcing ink through open mesh pores onto the substrate.

Prepress to Press Progression Animation



Visualizes digital design files transferring to mechanical press production.

Course Calculators

Self-Learning & Credit Calculator

Compute credits from scheme L:T:P:S hours.

Lecture Hours (L)

Tutorial Hours (T)

Practical Hours (P)

Credits = L + T + 0.5(P)

Screen Mesh Count Estimator

Recommend mesh count (threads/cm) by job type.

Job Substrate & Type

Select job type to see recommended mesh count.

WORKFLOW WALKTHROUGHS

Practical Step-by-Step Industry Walkthroughs

Walkthrough 1: Complete Prepress-to-Postpress Production Sequence

1. **Graphic Design:**

   Create vector graphics and layout in publishing software.

2. **Colour Separation:**

   Separate artwork into CMYK process color channels.

3. **Plate/Stencil Prep:**

   Output film positives or image CTP plates.

4. **Press Setup:**

   Mount plates/screens, meter ink, set registration guides.

5. **Printing Pass:**

   Run press, monitor ink density and registration.

6. **Finishing:**

   Cut to final size, fold, bind, and inspect quality.

Walkthrough 2: Two-Colour Screen Printing Registration

1. **Film Alignment:**

   Print registration crosshairs on artwork corners.

2. **First Screen Exposure:**

   Expose Screen 1 (Base Colour) using Positive 1.

3. **Second Screen Exposure:**

   Expose Screen 2 (Top Colour) using Positive 2.

4. **Bed Alignment:**

   Align Screen 1 to registration marks on bed lay-stops.

5. **First Pass:**

   Print Colour 1 and dry.

6. **Second Pass:**

   Lock Screen 2 into lay-stops, match crosshairs, print

Colour
2.

PRACTICAL ACTIVITY CENTRE

Laboratory Discipline, Safety & Practical Exercises

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: The official SITTTR experiment job list for 1101 (3 practical hours/week) is pending syllabus page release. Practical exercises will be updated upon receipt of official job lists.

Essential Laboratory Conduct Checklist

- [x] Wear lab apron and safety leather footwear.
- [x] Maintain clean work tables and degreased screens.
- [x] Store solvents in labeled, fire-safe containers.
- [x] Work collaboratively in assigned lab teams.

Teamwork & Laboratory Discipline (CO3/CO4)

Screen printing relies on disciplined teamwork: one student manages substrate feeding, one operates the squeegee, and one transfers printed sheets to drying racks while observing safety protocols.

EXPECTED PRACTICE QUESTIONS

Syllabus-Based Practice Questions with Answers

Question Bank Declaration: These expected practice questions are compiled strictly from confirmed Objectives 1–6 and Course Outcomes CO1–CO4. They are practice aids, not official SITTTR exam papers.

Objective / 1-Mark Questions

1. Define Prepress.
2. What does CMYK stand for?
3. Name the tool used to push ink through a screen mesh.
4. State the primary function of Postpress.
5. What material is commonly used for modern screen mesh?
6. Name one application of screen printing in electronics.
7. What is registration in multi-colour printing?
8. State the Bloom level for CO4.

View Answers

1. Activities prior to printing (design, plate/stencil prep);
2. Cyan, Magenta, Yellow, Key (Black);
3. Squeegee;
4. Finishing & binding operations;
5. Synthetic polyester;
6. Printed Circuit Board (PCB) resist printing;
7. Precise alignment of colours/images;
8. Apply.

Short Answer (3 Marks) Questions

1. Explain the role of printing in packaging and logistics.
2. Differentiate between Relief and Planographic printing.
3. List three mandatory laboratory safety rules during screen printing.
4. Describe the function of photo emulsion in stencil making.

View Model Points

1. Protects goods, provides barcodes/shipping labels, displays brand info.
2. Relief has raised image areas; Planographic is flat using oil/water repulsion.
3. Wear apron/shoes, ensure solvent ventilation, keep long hair tied near machines.
4. Light-sensitive coating that hardens under UV, leaving unexposed stencil areas open for ink.

Detailed / 7-Marks Questions

1. Explain the complete printing workflow across Prepress, Press, and Postpress stages.
2. Describe the step-by-step procedure for preparing a screen printing stencil and performing single-colour printing.

View Model Points

1. Prepress: Graphic design, CMYK separation, imposition, plate/stencil prep. Press: Mounting carrier, ink application, impression cylinder pass onto substrate. Postpress: Cutting, folding, stitching, trimming, quality inspection.

2. Steps: Degrease mesh → Coat emulsion → Dry in dark → UV exposure with film positive → Water washout stencil → Clamp in bed → Align registration → Squeegee ink stroke → Dry print → Clean screen.

PRACTICE MODEL PAPER

Sample Examination Paper — Based on Teaching & Assessment Scheme

Course 1101 — Introduction to Printing Technology

Time: 2½ Hours | Maximum: 60 Marks (ESE Theory Scheme)

Part A — Answer all questions (8 × 1 = 8 Marks)

1	Define Prepress workflow.	**1**
2	Name four process colors in CMYK.	**1**
3	Identify the image carrier in Screen Printing.	**1**
4	Define Registration in printing.	**1**
5	State one function of a Squeegee.	**1**
6	Name one Postpress finishing operation.	**1**
7	State one safety precaution when handling lab solvents.	**1**
8	State the total contact hours for Course 1101.	**1**

Part B — Answer short questions (8 × 3 = 24 Marks)

9. Explain the role of printing in daily life and education.
10. Describe the main activities involved in Prepress.
11. Compare Offset printing and Screen printing.
12. Explain the importance of laboratory discipline and safety.
13. Describe photo emulsion coating on screen mesh.
14. Explain how multi-colour registration is achieved in screen printing.

Part C — Answer detailed questions (4 × 7 = 28 Marks)

15A. Describe the sequential printing workflow from Prepress to Postpress with a neat flowchart. **OR 15B.** Explain major printing processes classified by image carrier surface.

16A. Describe step-by-step screen stencil preparation and squeegee printing operations. **OR 16B.** Detail laboratory safety rules, chemical handling, and emergency precautions for printing labs.

MASTER ANSWER KEY

Answer Direction for Practice Model Paper

1. Activities prior to printing (design, layout, plate/stencil prep).

2. Cyan, Magenta, Yellow, Black (Key).

3. Stencil on porous polyester mesh screen.

4. Precise alignment of images/colours on substrate.

5. Pushes and forces ink through open mesh pores.

6. Cutting / Folding / Binding / Lamination.

7. Ensure proper room ventilation and wear gloves/apron.

8. 90 Hours.

QUICK REVISION

Last-Day Summary Cards

Role of Printing

- Communication & media
- Packaging & labeling
- Advertising & logistics

Workflow Stages

- **Prepress:** Design & Plate
- **Press:** Ink Transfer
- **Postpress:** Cut & Bind

Printing Processes

- Relief (Flexo)
- Planographic (Offset)
- Intaglio (Gravure)
- Stencil (Screen)

Screen Lab Safety

- Apron & safety footwear
- Solvent ventilation
- Registration stops

BILINGUAL GLOSSARY

Printing Technology Technical Terms (English & Malayalam)

Printing Terminology Summary

English Technical Term	Malayalam Explanation (മലയാളം വിശദീകരണം)	Industry Definition
Prepress	പ്രിന്റിംഗിന് മുൻപുള്ള പ്രവർത്തനങ്ങൾ (ഡിസൈൻ, പ്ലേറ്റ് മേക്കിംഗ്)	All preparation steps performed prior to printing on press.
Press	അച്ചടി യന്ത്രത്തിലെ മഷി പകരൽ ഘട്ടം	The actual printing phase where ink is transferred to a substrate.
Postpress	അച്ചടിക്ക് ശേഷമുള്ള ഫിനിഷിംഗ് (കട്ടിംഗ്, ബൈൻഡിംഗ്)	Finishing operations such as cutting, folding, binding, and trimming.
Screen Printing	മെഷ് സ്ക്രീൻ വഴിയുള്ള അച്ചടി രീതി	Stencil printing process using a squeegee to force ink through mesh.
Squeegee	മഷി ഉരച്ചു നൽകുന്ന റബ്ബർ ഉപകരണം	Rubber or polyurethane blade used to push ink across the screen.
Mesh	സ്ക്രീൻ ഫ്രെയിമിലെ പോളിയെസ്റ്റർ വല	Woven synthetic fabric stretched tightly across a screen frame.
Registration	വർണ്ണങ്ങളുടെ കൃത്യമായ അലൈൻമെന്റ്	Precise positioning of print images during multi-colour printing.
Emulsion	പ്രകാശ സംവേദനത്വമുള്ള സ്ക്രീൻ കോട്ടിംഗ്	Light-sensitive chemical coating used to create screen stencils.
CMYK	നാല് പ്രക്രിയ വർണ്ണങ്ങൾ (സയാൻ, മജന്ത, മഞ്ഞ, കറുപ്പ്)	Cyan, Magenta, Yellow, and Key (Black) process colors.
Substrate	അച്ചടിക്കുന്ന പ്രതലം (പേപ്പർ, തുണി, പ്ലാസ്റ്റിക്)	Any material (paper, cardboard, fabric, plastic) being printed upon.

LEARNING RESOURCES & REFERENCES

Official Syllabus Reference Declaration

Not specified in the supplied official syllabus PDF — pending reference pages: Prescribed textbooks, digital learning resources, and official reference books for Course 1101 are pending syllabus page release.

Official Scope Declaration: Academic scope is strictly governed by the official SITTTR Kerala Diploma Revision 2026 Course Description for Subject Code 1101.